

# MATH 141B: Calculus with Applications to Biology II - Fall 2026

## Fall 2026 Syllabus

This PDF is a snapshot of the course website as of September 10, 2026. The website is the official, current version of the syllabus: <https://rfd131.github.io/syllabi/fa26/math141b/>

Some parts of the syllabus are only available online:

- [Quiz schedule](#) (which Learning Targets are offered each week)
- [Detailed learning objectives](#) for every Learning Target
- [Quiz session summary](#) and the live office-hours calendar
- [CourseHub login](#) for quiz sign-ups, progress reports, and course resources

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## Welcome to MATH 141B!

Welcome to Calculus with Applications to Biology II. Throughout the semester we will focus on developing and understanding the tools of calculus and applying these tools to problems in biology.

**Course Format:** The Fall 2026 semester runs as a "3+1" hybrid course with 3 credits of in-person instruction (MWF) and 1 WEB credit corresponding to online asynchronous content.

**Specifications Grading:** This course uses a specifications grading scheme. Your grade is NOT based on points or percentages, but on proficiency in learning objectives assessed through Learning Target Quizzes.

## Course Overview

### Prerequisites

- Math 140 or Math 140B, or equivalent.

### Class Structure

- **Monday, Wednesday, Friday:** Interactive in-class sessions with direct instruction, group problem-solving, and review.
- **Tuesday Recitations:** Required weekly Learning Target Quiz sessions. This may be at a different time than your MWF class time.
- **Online Components:** Interactive online textbook, Pre-lecture assignments (PLAs), video lecture content, online homeworks, practice problem assignments, and other course resources.

## Assessment Components

- **25 Learning Targets** assessed through weekly quiz sessions and exams
- **Experience Points (XP)** earned through:
  - Participation in LA Community Learning Sessions
  - Online homework (weekly)
  - Pre-lecture assignments (PLAs)
  - Online Practice Problems
  - Surveys and other assignments

**Note:** There are over 800 XP available throughout the semester. XP can modify your final grade through earned modifications.

## General Education Objectives

MATH 141B is part of Penn State's **General Education** curriculum and addresses the following general education objectives:

### Key Literacies

The ability to identify, interpret, create, communicate and compute using materials in a variety of media and contexts. Literacy acquired in multiple areas, such as textual, quantitative, information/technology, health, intercultural, historical, aesthetic, linguistic (world languages), and scientific, enables individuals to achieve their goals, to develop their knowledge and potential, to lead healthy and productive lives, and to participate fully in their community and wider society.

### Critical and Analytical Thinking

The habit of mind characterized by comprehensive exploration of issues, ideas, artifacts, and events before accepting or formulating a conclusion. It is the intellectually disciplined process of conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action.

## Course Instructors

### Max Kreider

*Instructor*

Pronouns: he/him/his

#### Teaching Section:

Section 1 - MWF 9:05 - 9:55 AM, 225 EE West

**Email:** [mbk6295@psu.edu](mailto:mbk6295@psu.edu)

**Office:** 230 McAllister

**Office Hours:**

- Tuesday 9:00 - 10:00 AM, 230 McAllister
- Wednesday 1:00 - 2:00 PM, 230 McAllister

[Shared](#)

*Other times by appointment; email to schedule.*

## Russ deForest

Pronouns: he/him

*Instructor, Course Coordinator*

### Teaching Section:

Section 3 - MWF 1:25 - 2:15 PM, 371 Willard

Email: [rfd131@psu.edu](mailto:rfd131@psu.edu)

Office: 005 McAllister

Office Hours:

- Monday 2:30 - 3:20 PM, 309 Sparks Shared
- Wednesday 12:20 - 1:10 PM, 103 EE West
- Friday 10:10 - 11:00 AM, 104 BBH

*Other times by appointment; email to schedule.*

## Steven Hair

Pronouns: he/him

*Instructor*

### Teaching Section:

Section 2 - MWF 12:20 - 1:10 PM, 011 Huck Life Sci

Email: [smh384@psu.edu](mailto:smh384@psu.edu)

Office: 104 McAllister

Office Hours:

- Wednesday 10:00 - 11:00 AM, 104 McAllister Shared
- Thursday 2:00 - 3:00 PM, 104 McAllister

*Other times by appointment; email to schedule.*

**Note:** Common Office Hours are shared office hour times open to students from any MATH 141B section. Find a time that works with your schedule. Also review your instructors full office hour schedule.

## Monday, Wednesday, Friday Classes

Classes meet three times per week for 50-minute sessions. Classes include interactive lectures, group problem-solving, and collaborative activities.

| Section   | Time            | Location          | Instructor    |
|-----------|-----------------|-------------------|---------------|
| Section 1 | 9:05 - 9:55 AM  | 225 EE West       | Max Kreider   |
| Section 2 | 12:20 - 1:10 PM | 011 Huck Life Sci | Steven Hair   |
| Section 3 | 1:25 - 2:15 PM  | 371 Willard       | Russ deForest |

## Tuesday Recitations - Learning Target Quiz Sessions

**Important:** Tuesday sessions are REQUIRED and are dedicated to Learning Target Quizzes. You must register for a specific Tuesday recitation section that may be held at a different time than your MWF class section.

Each Tuesday session is a 1 hour assessment period where you will take Learning Target Quizzes. These are low-stakes assessments with multiple opportunities to demonstrate proficiency throughout the semester.

| Section    | Time             | Location    |
|------------|------------------|-------------|
| Section 1R | 10:45 - 11:45 AM | 220 Hammond |
| Section 2R | 4:45 - 5:45 PM   | 101 Thomas  |

## Thursday Quiz Hours

Thursday Quiz Hours are additional sessions available for Learning Targets that you have not yet been successful in during a Tuesday or midterm exam session. **Thursday Quiz Hours begin in Week 4.** Learning Targets become available in quiz hours after they have been available at least twice in either a Tuesday session or on a Midterm Exam. LTs remain available during Thursday quiz hours for the rest of the semester according to a schedule.

**12:15 - 1:15 PM**

26 Hosler

**2:15 - 3:15 PM**

112 Kern

**4:45 - 5:45 PM**

101 Thomas

**Note:** It is not required that you attend Thursday Quiz Hours, but most students will use quiz hours at least a few times during the semester and you should plan space in your schedule.

**Note:** Sign up for Quiz Hours through the same quiz sign up form you use for Tuesday sessions.

## Course Materials

We provide an interactive online textbook that includes video lectures, class handouts, sample quizzes, practice problems, and outside of class assignments. It may still be helpful to have a calculus textbook as a reference and a source of additional practice problems.

### CourseHub Materials

The following materials are provided through our online textbook in the CourseHub web app:

- Video lectures for each topic
- Pre-lecture assignments (PLAs), including video lectures, due before each class
- Online homework problems (due weekly on Mondays)
- Sample Quiz and Study Guide for each Learning Target with additional practice problems and links to textbook examples
- In-class activities, with solutions posted in the textbook after class
- Interactives and tutorials
- Online practice problems for each Learning Target

**Access CourseHub:** [CourseHub Login](#)

### Free Online Textbooks

**Differential Calculus for the Life Sciences** FREE

by Leah Edelstein-Keshet (2019)

An excellent open source textbook that overlaps significantly with our course but does not include all topics that we will cover. This is a great source of practice problems and interesting applications. Available both as a PDF and through LibreText.

[Download PDF](#)[View on LibreText](#)

## Calculus FREE

by Gilbert Strang, Wellesley-Cambridge Press (1991)

Available through MIT OpenCourseWare. The first 7 chapters of this text overlap with topics we cover in MATH 141B. It is an excellent source of problems for additional practice. It does not focus on biology, but does include some biological applications. You can download a PDF of this text at the link below.

[Download PDF](#)[View on MIT OCW](#)

## Optional Hardcover Textbooks

If you prefer to have a hardcover Calculus text for reference, the following textbooks available through online sellers are all good choices for reference or as a source for more practice problems:

**Calculus for Biology and Medicine** by Claudia Neuhauser  
*Comprehensive coverage with biological applications throughout*

**Modeling the Dynamics of Life** by Fred Adler  
*A primary source for our online text, with extensive biological modeling*

**Calculus for the Life Sciences** by Sebastian Schreiber et al.  
*Includes core Calculus I content and many biological examples*

## Learning Targets Overview

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There are **25 Learning Targets** in MATH 141B. You are assessed on these learning targets through weekly "Learning Target Quiz Sessions" every Tuesday beginning in Week 2. This page provides a comprehensive overview of the Learning Target system and how it works.

[View All Learning Targets with Descriptions](#)

## How Learning Target Quizzes Work

### Grading System

Learning Target Quizzes are graded on a **Success/Not Yet** basis:

**Success!** Solutions for all problems are free from conceptual errors and contain few or no computational errors.

**Not Yet** The quiz attempt shows that additional skill development or conceptual understanding is needed. This is an opportunity to learn and improve!

## Quiz Types

There are three main types of Learning Target quizzes:

**Foundational (F)** Cover introductory concepts and foundational skills for a related set of learning objectives.

**Advanced (Adv)** Cover advanced concepts and skills, often requiring application of foundational skills to problems.

**One-Time Proficiency** Counted as both Complete and Proficient after one success. May involve both foundational and advanced skills.

**Group** Several LTs throughout the semester will be assessed as group quizzes, where you will work with your peers.

### 1. Success on one part

(F or Adv)  
Earns Complete

### 2. Success in all parts

(F and Adv)  
Earns Proficient

### 3. Grade Impact

Number Complete and  
Number Proficient both  
count toward final grade

## Achievement Levels

**Complete** You earn this when you get a **Success!** on either the Foundational (F) or Advanced (Adv) quiz version. Once you earn a Success in an LT Quiz Version, you do not attempt that version again.

**Proficient** You earn this when you get a **Success!** on both the Foundational (F) and Advanced (Adv) quiz versions for the same Learning Target.

**Important:** You don't need to earn a success in the Foundational version before attempting the Advanced version. You can attempt them in any order.

**Note:** For One-Time Proficiency targets, one success counts as both Complete and Proficient.

## Learning Target Quiz Sessions

Quiz sessions are held **every Tuesday** throughout the semester. You will typically take several short quizzes during a session, with each quiz focusing on a single Learning Target.

**Important:** Participation in the Learning Target Quiz Sessions each week is **required** and is **essential** to your success. Most learning targets will be available 4-5 times throughout the semester according to a schedule posted in CourseHub and in the syllabus.

## Retaking Learning Target Quizzes

If a learning target quiz is marked **Not Yet**, you'll have another opportunity to earn a **Success!** on that target. Use the feedback provided on Gradescope to understand what material needs further review before re-attempting.

**Important Requirement:** We require you to attempt at least one of the newest LT quizzes that become available each week, even if you feel less than fully prepared. This helps maintain momentum and provides valuable practice.

**For the foundational targets LT1F through LT9F:** If you are not able to earn a success on these targets in the first two times they are available, we will require you to visit an instructor office hour for help reviewing important ideas so that you can be better prepared to succeed on the next attempt.

## Resources by Learning Target

You'll find resources organized by learning target in the online textbook where you can review video lectures, complete online practice problems, and try sample quizzes for each LT quiz. You can also practice quiz content and work with peers at LA Evening Community Learning Sessions throughout the semester.

## Frequently Asked Questions

### Q: How many times can I attempt each Learning Target?

Most learning targets will be available 4-5 times throughout the semester according to the posted schedule. You can attempt each LT Quiz version each time it's offered until you achieve Success. However, if you do not earn Success, don't wait to seek out help! Use each attempt and the feedback to improve your understanding.

### Q: What happens if I miss a Tuesday quiz session?

Missing quiz sessions will limit your opportunities to demonstrate proficiency on learning targets. While missing 1-2 sessions may not greatly impact your grade, regular attendance is essential. Missing more than 2-3 sessions will likely be detrimental to your grade. See the quiz make up policy under Course Policies.

### Q: Do I need to get a "perfect score" on a Learning Target quiz to earn a Success?

No! Earning a Success means you've demonstrated proficiency and understanding in the skills for that Learning Target. While it may feel like you need "100%", this is different from a traditional exam setting. Each Learning Target Quiz assesses a small, focused set of skills. You get feedback after each attempt to help you improve.

**Q: Do I need to complete the Learning Targets in order?**

No, you don't need to earn Success on LT1F before attempting LT2F or LT3F and you do not need to earn a Success on LT1F before attempting LT1Adv. While some learning targets build on skills from earlier targets and the material is interconnected, you should continue attempting multiple learning target quizzes each week to keep pace with the course.

**Q: Can I attempt the Advanced version before the Foundational version?**

Yes! You don't need to earn a success in the Foundational version before attempting the Advanced version. For each target, the Advanced (Adv) version becomes available 1-2 weeks after the Foundational (F) version is first available.

**Q: Is "Not Yet" on a Learning Target Quiz the same as failing?**

"Not Yet" is intended to emphasize that you can revisit the skills and come back for another attempt. Your final grade depends on the number of Learning Targets you successfully complete and achieve proficiency in by the end of the semester.

**Q: How do exams relate to Learning Target Quizzes?**

Exams provide additional opportunities to earn Complete or Proficient status on learning targets. Each exam includes a set of active learning targets, and you choose which targets to focus on during the exam.

**Q: Will there be additional quiz opportunities outside of Tuesday sessions?**

Yes, beginning in Week 4, we'll offer proctored Quiz Hours for students who need to attempt more quizzes than can reasonably fit in a single Tuesday session. These are on Thursdays. The times and locations are posted in this syllabus and in CourseHub.

**Q: What resources can I use during Learning Target Quizzes?**

The LT Quizzes are proctored quizzes taken with pencil and paper. Notes, calculators, phones, and electronic devices are not permitted during the quiz sessions. These are closed-book assessments designed to assess your understanding of the material.

**[View All Learning Targets with Descriptions](#)**



# MATH 141B Learning Targets

This page lists all 25 Learning Targets for MATH 141B. Detailed learning objectives for every target, including the Foundational and Advanced versions of two-part targets, are on the website: <https://rfd131.github.io/syllabi/fa26/math141b/learning-targets-list.html>.

## Learning Target Types

**Two-Part** Foundational (F) + Advanced (Adv) versions    **One-Time** One success = Complete + Proficient  
**Group** Assessed as group quizzes

|      |          |                                             |
|------|----------|---------------------------------------------|
| LT1  | Two-Part | Review topics from Calculus I               |
| LT2  | Two-Part | Integration by Substitution                 |
| LT3  | Two-Part | Integration by Parts                        |
| LT4  | Two-Part | Integration by Partial Fractions            |
| LT5  | One-Time | Integration Techniques                      |
| LT6  | Two-Part | Modeling with Differential Equations        |
| LT7  | Two-Part | Separation of Variables                     |
| LT8  | Two-Part | The Integrating Factor Method               |
| LT9  | Two-Part | Linear Models in Biology                    |
| LT10 | Two-Part | Slope Fields                                |
| LT11 | Two-Part | Phase Lines and Equilibrium Solutions       |
| LT12 | Group    | Bifurcation Diagrams                        |
| LT13 | One-Time | Integration Computation and Applications    |
| LT14 | Two-Part | Vector Operations                           |
| LT15 | Two-Part | Linear Transformations                      |
| LT16 | One-Time | Matrix Operations                           |
| LT17 | Two-Part | Discrete Dynamical Systems                  |
| LT18 | Two-Part | The Inverse Matrix                          |
| LT19 | Two-Part | Gaussian Elimination                        |
| LT20 | Two-Part | Eigenvalues and Eigenvectors                |
| LT21 | One-Time | Solving Discrete Time Matrix Models         |
| LT22 | One-Time | Markov Models                               |
| LT23 | One-Time | Interpreting Discrete Time Matrix Models    |
| LT24 | One-Time | Linear Systems of Differential Equations    |
| LT25 | Group    | Nonlinear Systems of Differential Equations |

## Grading Overview

**Specifications Grading:** Your grade is NOT based on points or percentages. Instead, your grade is determined by demonstrating proficiency in learning objectives through Learning Target Quizzes, with modifications possible through Experience Points (XP).

### Your final grade is determined by:

- The number of **Learning Targets** marked as **Complete** (at least one part successful)
- The number of **Learning Targets** marked as **Proficient** (all parts successful)
- Experience points (XP) earned through online work and participation

**Important:** You do not lose points by attempting assignments. Every assignment is a learning opportunity!

## How Your Grade is Determined

### MATH 141B (4 credits)

| Letter Grade | Learning Targets Complete<br>(Success on one part) | Learning Targets Proficient<br>(Success on all parts) |
|--------------|----------------------------------------------------|-------------------------------------------------------|
| <b>A</b>     | 25                                                 | 21                                                    |
| <b>A-</b>    | 23                                                 | 19                                                    |
| <b>B+</b>    | 22                                                 | 17                                                    |
| <b>B</b>     | 21                                                 | 15                                                    |
| <b>B-</b>    | 20                                                 | 13                                                    |
| <b>C+</b>    | 19                                                 | 11                                                    |
| <b>C</b>     | 18                                                 | 9                                                     |
| <b>D</b>     | 14                                                 | —                                                     |
| <b>F</b>     | If requirements for D are not met                  |                                                       |

### Important Note

You must meet ALL requirements to receive the corresponding letter grade. For example:

- If you have 25 Complete targets and 15 Proficient targets, your final grade would be **B** (not A)
- If you have 23 Complete targets and 17 Proficient targets, your final grade would be **B+** (not A-)

Modifications earned through XP can help you meet these thresholds (see below).

## Experience Points (XP)

### Earning Experience Points

Course engagement at a satisfactory level typically corresponds to earning about 500-600 XP (enough for 2 modifications).

| Activity                              | XP Value             | Details                                                                                                               |
|---------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>LA Community Learning Sessions</b> | 20 XP per week       | Attend and participate in weekly LA sessions                                                                          |
| <b>Pre-lecture Assignments (PLAs)</b> | 5 XP per PLA         | Complete before each MWF class. XP is awarded 40% (2/5) for engagement and 60% (3/5) for correctness on the problems. |
| <b>Online Homework</b>                | 10 XP per assignment | Due weekly on Mondays. XP score based on percentage of available points.                                              |
| <b>Target Practice</b>                | 5 XP                 | Available for each Learning Target. XP score based on percentage of available points.                                 |
| <b>Quiz sign-ups</b>                  | 2 XP                 | Sign up each week for the quiz sessions you plan to attend.                                                           |
| <b>Surveys and other assignments</b>  | Varies by assignment | Occasional surveys and other assignments posted on CourseHub.                                                         |

## XP Modifications

### How Modifications Work

The number of learning targets Complete or Proficient may be modified by experience points (XP) earned during the course.

**Each modification** earned will:

- Increase the number of Complete targets by one, **OR**
- Increase the number of Proficient targets by one.

| XP Earned        | Effect                    |
|------------------|---------------------------|
| Less than 250 XP | No modifications          |
| 250-499 XP       | One modification          |
| 500-749 XP       | Two modifications         |
| 750 or more XP   | 3 modifications (maximum) |

**Important:** Modifications are automatically applied at the end of the course in the way that will best benefit your final grade.

### Example

A student finishes the semester with:

- 23 Complete Learning Targets
- 17 Proficient Learning Targets
- 550 XP earned (= 2 modifications)

The 2 modifications could be applied to increase Proficient targets from 17 to 19, resulting in a final grade of **A-** (meeting the requirements of 23 Complete and 19 Proficient).

**Note:** Modifications are NOT applied to specific targets. For example, you cannot use one modification on a One-Time Proficiency target to increase both Complete and Proficient counts by one.

## Understanding Learning Target Achievement

### Complete vs. Proficient

#### To Complete a Learning Target:

- Earn a Success on either the Foundational (F) Quiz Version OR the Advanced (Adv) Quiz Version
- For One-Time Proficiency Targets, one success counts as both Complete and Proficient
- Once you earn a Success in an LT Quiz Version, you do not attempt that version again

#### To be Proficient in a Learning Target:

- Earn a Success on BOTH the Foundational (F) Quiz Version AND the Advanced (Adv) Quiz Version
- For One-Time Proficiency Targets, one success counts as both Complete and Proficient
- You don't need to earn a success in the Foundational version before attempting the Advanced version

See the [Learning Targets page](#) for more details about the quiz system.

## Course Policies

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### Learning Target Quiz Session Policies

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#### Tuesday Quiz Sessions (Recitations)

Learning Target Quizzes are held on **Tuesdays** throughout the semester beginning in Week 2. These sessions are **required** and are the primary way you demonstrate proficiency on Learning Targets.

### **Tuesday Quiz Session Policies:**

- You will submit a form available through CourseHub indicating the LT quizzes you wish to attempt in the session (Due Sunday night before the Tuesday session).
- If you do not submit the quiz sign up form you should still attend your Tuesday recitation. A packet will be printed for you, but without signing up, you cannot be certain that it includes the LT Quizzes you wish to attempt that week.
- You must attend the Tuesday recitation you are enrolled in.
- To facilitate efficient distribution of the quiz packets, you will be seated according to a seating chart that is established in the Week 1 session.
- When you arrive at the Tuesday Quiz Session, you will leave bags/backpacks at the front of the room and take your seat according to the seating chart. Phones, calculators, notes, smart watches, electronic devices, or any study aids are not permitted during the quiz session. At your seat, you should need only pencil(s)/pen(s) and your student id. Your phone should be left in your bag.
- When you are done with your quiz packet, you will turn the quiz packet in to one of the proctors and show your id.
- If you complete your quiz packet with 15 minutes or more remaining in the session, you may request a copy of an LT quiz that was not part of your packet.
- Any quizzes or quiz packets distributed during the session must be turned in regardless of the status of completion.
- When you have turned in your quiz packet you may retrieve your things and leave the session. If you wish to remain in the session, you will need to sit quietly and retrieve your things at the end of the session.

## Thursday Quiz Hours

### Thursday Quiz Hour Policies:

- You will submit a form through Course Hub indicating the LT quizzes you wish to attempt in the session (Due Wednesday at 5 PM the day before the session).
- The Thursday quiz hour sessions are optional. If you do not sign up in advance no packet will be printed for you.
- You must attend the Thursday quiz hour you signed up for.
- When you arrive at the Thursday quiz hour you will leave bags/backpacks at the front of the room and get your quiz packet from the proctors. Quiz packets are in alphabetical order by first name. Phones, calculators, notes, smart watches, electronic devices, or any study aids are not permitted during the quiz session. At your seat, you should need only pencil(s)/pen(s). Your phone should be left in your bag. If your id is on your phone, you will retrieve your phone when you turn in your packet. You cannot have your phone with you while you take the quiz.
- When you are done with your quiz packet, you will turn the quiz packet in to one of the proctors and show your id.
- If you complete your quiz packet with 15 minutes or more remaining in the session, you may request a copy of an LT quiz that was not part of your packet.
- Any quizzes or quiz packets distributed during the session must be turned in regardless of the status of completion.
- When you have turned in your quiz packet you may retrieve your things and leave the session. If you wish to remain in the session, you will need to sit quietly and retrieve your things at the end of the session.

## Make-up Quiz Sessions

We understand that sometimes you may need to miss a Tuesday quiz session due to illness or other unavoidable circumstances. LTs that you miss on one particular Tuesday will still be available at a later time. In addition, we offer **two designated make-up quiz sessions during the semester**. This provides a fair opportunity to make-up missed LT quizzes for students who miss up to 2-3 Tuesday quiz sessions during the semester. We cannot accommodate students who routinely and repeatedly miss Tuesday quiz sessions.

### Semester Make-up Quiz Sessions

#### Make up Quiz Session 1

Date: 10/21/2026, 4:45 - 8:30 PM

Location: 112 Forest Resources Bldg

Week 9, Make up for Tues Quiz Sessions, Week 2 through Week 8

#### Make up Quiz Session 2

Date: 12/2/2026, 4:45 - 8:30 PM

Location: 112 Forest Resources Bldg

Week 14, Make up for Tues Quiz Sessions, Week 8 through Week 13

### Important Make-up Quiz Details:

- You can attempt Learning Targets from the Tuesday session(s) you missed (maximum of 6 LTs).
- You may not need the make-up session by the time it becomes available. For example, if you miss the Week 2 quiz session, when LT1F, LT2F, and LT3F are available and you succeed on these LTs before the make-up, you will not need the make-up session for Week 2.
- These make-ups accommodate students who miss 2-3 Tuesday sessions during the semester. We cannot accommodate students who routinely or repeatedly miss Tuesday quiz sessions.
- Sign-up is required. Forms will be posted in CourseHub

## Exams

We have two midterm exams and a final exam in MATH 141B. Each exam provides additional opportunities to attempt Learning Targets.

### Midterm Exams

Midterm exams are important opportunities to succeed on a larger number of LTs than are available in a typical Tuesday quiz session.

#### Midterm One

**Regular Exam:** 9/22/2026  
**Time:** 6:15 - 7:30 PM  
**Location:** varies by instructor

**Conflict/Make-up:** 9/24/2026  
**Time:** 4:45 - 8:30 PM  
**Location:** 112 Chambers

#### Midterm Two

**Regular Exam:** 10/27/2026  
**Time:** 6:15 - 7:30 PM  
**Location:** varies by instructor

**Conflict/Make-up:** 10/29/2026  
**Time:** 4:45 - 8:30 PM  
**Location:** 112 Chambers

#### Midterm Conflict and Make-up Policy

- If you have an academic conflict during the scheduled evening midterm or need to make up the midterm due to a valid reason, we will schedule a make-up exam.
- The scheduled conflict/make-up opportunity takes place on Thursday during the same week as the regularly scheduled exam.

- Students who are not able to use this scheduled make-up opportunity will be scheduled for a make-up individually. This make-up opportunity must be completed within two weeks of the regularly scheduled exam.

## Final Exam

### Important Final Exam Information:

- Your final exam will consist of any learning targets that you have not yet earned a Success on.
- Prior to finals week, your Progress Report will indicate your course letter grade without the final exam.
- If you meet the requirements for an A in the course prior to the final, no final exam will be printed for you.
- The final exam is your last opportunity to demonstrate proficiency on any remaining Learning Targets.

**Final Exam Date:** To be announced (Check the Penn State Final Exam Schedule)

**Important Travel Policy:** Personal travel plans (including early departure for breaks) are NOT considered excused absences for quizzes or exams. Please plan your travel accordingly.

## Academic Integrity

All [Penn State Policies](#) regarding ethics and academic integrity apply to this course. Academic integrity is the pursuit of scholarly activity free from fraud and deception and is an educational objective of this institution.

Academic dishonesty includes, but is not limited to:

- Cheating
- Plagiarizing
- Fabricating of information or citations
- Facilitating acts of academic dishonesty by others
- Having unauthorized possession of examinations
- Submitting work of another person or work previously used without informing the instructor
- Tampering with the academic work of other students

For any material or ideas obtained from other sources, such as text or things you see on the web, in the library, etc., a source reference must be given. Direct quotes from any source must be identified as such. All exam answers must be your own, and you must not provide any assistance to other students during exams.

**Any instances of academic dishonesty WILL be pursued under the University and Eberly College of Science regulations concerning academic integrity.**



## Generative AI Policy

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**AI as a Learning Tool:** When used appropriately, generative AI tools (such as ChatGPT, Claude, Copilot, etc.) can be powerful resources to enhance your learning experience. We encourage you to explore these tools in ways that support genuine understanding and skill development. Instructors may periodically discuss effective strategies for using AI to strengthen your mathematical understanding.

To maintain academic integrity while leveraging AI's benefits, please follow these guidelines:

### Permitted Uses

- Clarifying mathematical concepts outside of class
- Generating additional practice problems for self-study
- Checking your understanding of definitions and theorems
- Getting help with study strategies
- Understanding error messages in approved software

### Not Permitted

- Using AI to complete graded assignments and as a substitute for authentic demonstrations of your understanding.
- Generating solutions for homework
- Accessing AI during any in-person assessments.

**Academic Integrity Violation:** Using generative AI to complete graded work constitutes academic dishonesty and will be treated as such. This includes both direct copying and substantial paraphrasing of AI-generated content.

**Why this policy?** Learning calculus requires developing your own problem-solving skills and mathematical intuition. Using AI to complete assignments prevents you from developing these essential skills and undermines the learning objectives of the course.

**If you're unsure:** Please ask your instructor before using AI tools for any course-related work. We're happy to clarify what is and isn't appropriate.

## Attendance Policy

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**In-Person Classes (MWF):** Regular attendance and participation in your MWF section is expected. Active participation and engagement during class contribute to your success and will help you become proficient in the material.

**Tuesday Learning Target Quiz Sessions:** Attendance is **REQUIRED**. These sessions are the primary way you demonstrate proficiency on Learning Targets, which directly determines your grade. Missing more than 2-3 sessions will likely be detrimental to your grade.

### Late Work Policy

**PLAs:** You start the semester with **10 PLA Late Passes**. A PLA late pass extends the PLA assignment until the next class period at 11:59 PM. To use one, open the late assignment and select "Use a late pass."

**Weekly Online Homeworks:** You start the semester with **2 Weekly Homework Late Passes**. A weekly homework late pass extends the assignment by one week — except the final homework of the semester, which can only be extended until Friday at 11:59 PM on the final class day.

Once a late pass extension has expired, these assignments remain available for ungraded practice but will not count toward XP. Late passes are used only for these online assignments and are tracked automatically in the online textbook.

### Disability Accommodation

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Penn State welcomes students with disabilities into the University's educational programs. Every Penn State campus has an office for students with disabilities. The Office for Disability Services (ODS) Web site provides contact information for every Penn State campus.

For further information, please visit the Office for Disability Services Web site: <http://equity.psu.edu/ods>

In order to receive consideration for reasonable accommodations, you must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: <https://equity.psu.edu/offices/student-disability-resources/documentation>

If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with an accommodation letter. Please share this letter with your instructors and discuss the accommodations with them as early as possible. You must follow this process for every semester that you request accommodations.

### Getting Help in MATH 141B

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Success in MATH 141B comes from actively engaging with the material and seeking help when needed. We offer multiple resources to support your learning throughout the semester. Don't wait to get help.

**Remember:** For the first 9 foundational learning targets (LT1F-LT9F), if you don't earn a Success during the first two attempts, you'll need to visit an instructor office hour before attempting again. This requirement helps ensure you get the support needed to succeed.

### Office Hours

Office hours are a time for one-on-one help. You can review course material, get clarification about quiz feedback, or discuss concerns about the course. Office hours are "drop-in". No appointment is needed for scheduled office hours.

## Common Office Hours

Common office hours are open to students from all sections, regardless of who teaches your section.

| Day       | Time             | Location       | Instructor    |
|-----------|------------------|----------------|---------------|
| Monday    | 2:30 - 3:20 PM   | 309 Sparks     | Russ deForest |
| Wednesday | 1:00 - 2:00 PM   | 230 McAllister | Max Kreider   |
| Wednesday | 10:00 - 11:00 AM | 104 McAllister | Steven Hair   |

## All Instructor Office Hours

### Max Kreider

**Tuesday:** 9:00 - 10:00 AM  
230 McAllister

**Wednesday:** 1:00 - 2:00 PM  
230 McAllister (*Shared*)

*Other times by appointment; use the Course Hub App to schedule.*

### Russ deForest

**Monday:** 2:30 - 3:20 PM  
309 Sparks (*Shared*)

**Wednesday:** 12:20 - 1:10 PM  
103 EE West

**Friday:** 10:10 - 11:00 AM  
104 BBH

*Other times by appointment; use the Course Hub App to schedule.*

### Steven Hair

**Wednesday:** 10:00 - 11:00 AM  
104 McAllister (*Shared*)

**Thursday:** 2:00 - 3:00 PM  
104 McAllister

*Other times by appointment; use the Course Hub App to schedule.*

## LA Community Learning Sessions

### Structured Group Learning with Learning Assistants

Learning Assistants run community learning sessions most weeks during the semester. These are structured problem-solving sessions where students work in groups reviewing recently covered class material.

#### What to Expect:

- Work in small groups with guidance and support from Course LAs
- Practice and review concepts and skills from recent classes
- Work on practice quizzes in preparation for upcoming quiz sessions
- Form peer study groups that extend beyond the classroom and LA Sessions
- Participation in LA Sessions earns 20 XP

#### Schedule:

**Location:** 201 Thomas Building

- **6:00 - 7:00 PM:** Wednesdays and Thursdays
- **7:00 - 8:00 PM:** Wednesdays and Thursdays

**Earn XP:** Participation in LA Community Learning Sessions earns 20 Experience Points per week, contributing to potential grade modifications.

## The Math Center @ Penn State Learning (Free Drop-in Tutoring)

### Drop-in Tutoring at 7 Sparks Building

The Math Center offers free drop-in tutoring where students work individually or in small groups with undergraduate peer tutors who have excelled in MATH 141B. Sessions are designed to address specific needs including homework help, quiz and exam preparation, and understanding particular concepts.

#### What to Expect:

- **Session Length:** Typically up to 30 minutes per student if there's a wait
- **Location:** 7 Sparks Building
- **Format:** Individual or small group tutoring
- **Come Prepared:** Bring questions and course materials

#### How Tutors Help:

- Clarify course concepts with alternative explanations
- Work through similar practice problems
- Review material from the week
- Address prerequisite knowledge gaps
- Help with quiz and exam preparation

## Tips for Getting the Most Out of Help Resources

- **Start early:** Don't wait to seek help.
- **Be consistent:** Regular in-class attendance and participation, consistent participation in LA Community Learning Sessions, and consistently working in study groups with your peers are more effective than cramming.
- **Form study groups:** Connect with classmates during class and in LA sessions.
- **Review quiz feedback:** Prepare for office hours by reviewing previous quiz attempts and preparing questions.
- **Ask questions in class:** Instructors and LAs are there to help. Class time is an opportunity to resolve misunderstandings.

## Student Resources

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Penn State offers a wide range of resources to support your academic success and personal well-being. This page provides important information about services available to all students.

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For further information, please visit the Office for Disability Services Web site: <http://equity.psu.edu/ods>

In order to receive consideration for reasonable accommodations, you must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: <http://equity.psu.edu/ods/doc-guidelines>

If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with an accommodation letter. Please share this letter with your instructors and discuss the accommodations with them as early as possible. You must follow this process for every semester that you request accommodations.

### Nondiscrimination Statement

The Pennsylvania State University is committed to and accountable for advancing diversity, equity, inclusion, and sustainability in all of its forms. We embrace individual uniqueness, foster a culture of inclusion that supports both broad and specific diversity initiatives, leverage the educational and institutional benefits of diversity, and engage all individuals to help them thrive. We value inclusion as a core strength and an essential element of our public service mission.

Penn State is an equal opportunity, affirmative action employer, and is committed to providing employment opportunities to all qualified applicants without regard to race, color, religion, age, sex, sexual orientation, gender identity, national origin, disability or protected veteran status.

## Educational Equity

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Penn State University has adopted a [Protocol for Responding to Bias Motivated Incidents](#) that is grounded in the policy that the University is committed to creating an educational environment which is free from intolerance directed toward individuals or groups and strives to create and maintain an environment that fosters respect for others.

Students, faculty, or staff who experience or witness a possible bias motivated incident are urged to report the incident immediately by doing one of the following:

- Submit a report via the Report Bias webpage: <http://equity.psu.edu/reportbias/>
- Contact one of the following offices:
  - University Police Services, University Park: 814-863-1111
  - Multicultural Resource Center, Diversity Advocate for Students: 814-865-1773
  - Office of the Vice Provost for Educational Equity: 814-865-5906
  - Office of the Vice President for Student Affairs: 814-865-0909
  - Affirmative Action Office: 814-863-0471

### Counseling and Psychological Services

Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The university offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings.

These services are provided by staff who welcome all students and embrace a philosophy respectful of clients' cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

- Counseling and Psychological Services at University Park (CAPS): <http://studentaffairs.psu.edu/counseling/> - 814-863-0395
- Penn State Crisis Line (24 hours/7 days/week): 877-229-6400
- Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741

## Academic Support Resources

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Penn State offers numerous resources to help you succeed academically:

- **Penn State Learning:** Free tutoring and guided study groups. Visit [pennstatelearning.psu.edu](http://pennstatelearning.psu.edu)
- **Library Resources:** Research assistance and study spaces. Visit [libraries.psu.edu](http://libraries.psu.edu)
- **Writing Center:** Help with writing assignments across all disciplines
- **Academic Success Center:** Study skills workshops and academic coaching

## Technology Resources

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Technical support and resources for students:

- **IT Service Desk:** Technical support for Penn State systems - 814-865-HELP (4357)
- **Canvas Support:** 24/7 support for the learning management system
- **Software Downloads:** Free and discounted software for students
- **Computer Labs:** Available across campus with specialized software